

Referencia para Reproducir un Modelo Lineal Desde Cero

Regresion lineal multiple con datos_5_ej5.csv

Material de estudio

Table of contents

1	Proposito	1
2	Datos	2
3	Ajuste de Referencia con <code>lm()</code>	3
4	Coeficientes e Inferencia Individual	4
5	Valores Ajustados y Residuos	5
6	Sumas de Cuadrados	6
7	R Cuadrada y Error Estandar Residual	7
8	Prueba F Global	8
9	Devianza, Log-Verosimilitud, AIC y BIC	9
10	Modelo Reducido y Prueba F Parcial	10
11	Prueba de Razon de Verosimilitudes	12
12	Objetivos para la Implementacion Manual	12
13	Referencias Numericas Compactas	13

1 Proposito

Este documento fija un modelo de referencia ajustado con `lm()` para comparar contra una implementacion manual posterior, por ejemplo en C++/Rcpp. El objetivo no es usar `lm()` como metodo de calculo final, sino como patron numerico confiable para verificar coeficientes, residuos, sumas de cuadrados, estadisticos, intervalos, p-values, devianza y criterios de informacion.

Usaremos un caso deliberadamente estandar: regresion lineal multiple con intercepto y cinco predictores numericos, sin factores ni interacciones.

El modelo completo es:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i4} + \beta_5 x_{i5} + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2).$$

2 Datos

```
datos <- read.csv("Clase/codes/datos_5_ej5.csv")
str(datos)
```

```
'data.frame':  20 obs. of  6 variables:
 $ x1: num  3.83 2.89 2.86 2.92 3.06 2.07 2.52 2.45 3.13 2.44 ...
 $ x2: num  28.9 20.1 69 65.4 29.6 ...
 $ x3: num  7.2 -11.71 12.32 14.28 6.31 ...
 $ x4: num  26.6 24.4 25.7 25.7 25.4 ...
 $ x5: num  6.19 5.17 7.04 7.1 6.15 6.41 6.86 5.78 6.51 5.57 ...
 $ y : num  37 26.5 36.5 40.7 37.1 ...
```

```
datos
```

	x1	x2	x3	x4	x5	y
1	3.83	28.87	7.20	26.60	6.19	37.01
2	2.89	20.10	-11.71	24.40	5.17	26.51
3	2.86	69.05	12.32	25.70	7.04	36.51
4	2.92	65.40	14.28	25.70	7.10	40.70
5	3.06	29.59	6.31	25.40	6.15	37.10
6	2.07	44.82	6.16	21.60	6.41	33.90
7	2.52	77.37	12.70	24.90	6.86	41.80
8	2.45	24.67	-0.17	25.01	5.78	33.40
9	3.13	65.01	9.85	26.60	6.51	41.01
10	2.44	9.99	-0.05	28.01	5.57	37.20
11	2.09	12.20	-12.86	23.51	5.62	23.30
12	2.52	22.55	0.92	23.60	5.34	35.20
13	2.22	14.30	4.77	24.51	5.80	34.90
14	2.67	31.79	-0.96	25.80	6.19	33.10
15	2.71	11.60	-16.04	25.20	5.62	22.70
16	3.14	68.47	10.62	25.01	6.94	39.70
17	3.54	42.64	2.66	25.01	6.33	31.80
18	2.52	16.70	-10.99	24.80	6.01	31.70
19	2.68	86.27	15.03	25.51	7.51	43.10
20	2.37	76.73	12.77	24.51	6.96	41.01

```
summary(datos)
```

x1		x2		x3		x4	
Min.	:2.070	Min.	: 9.99	Min.	:-16.0400	Min.	:21.60
1st Qu.:	:2.447	1st Qu.:	:19.25	1st Qu.:	-0.3675	1st Qu.:	:24.51
Median	:2.675	Median	:30.69	Median	: 5.4650	Median	:25.01
Mean	:2.732	Mean	:40.91	Mean	: 3.1405	Mean	:25.07
3rd Qu.:	:2.955	3rd Qu.:	:66.17	3rd Qu.:	:11.0450	3rd Qu.:	:25.70
Max.	:3.830	Max.	:86.27	Max.	:15.0300	Max.	:28.01

x5		y	
Min.	:5.170	Min.	:22.70
1st Qu.:	:5.740	1st Qu.:	:32.77
Median	:6.190	Median	:35.85
Mean	:6.255	Mean	:35.08
3rd Qu.:	:6.880	3rd Qu.:	:39.95
Max.	:7.510	Max.	:43.10

3 Ajuste de Referencia con lm()

```
modelo_ref <- lm(y ~ x1 + x2 + x3 + x4 + x5, data = datos)
summary(modelo_ref)
```

Call:

```
lm(formula = y ~ x1 + x2 + x3 + x4 + x5, data = datos)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.9497	-0.6174	0.0623	0.7343	5.0018

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	19.94857	13.62755	1.464	0.1653
x1	-1.79333	1.23340	-1.454	0.1680
x2	0.04360	0.05326	0.819	0.4267
x3	0.55576	0.09296	5.979	3.38e-05 ***
x4	1.11017	0.43377	2.559	0.0227 *
x5	-1.81092	2.02739	-0.893	0.3868

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.074 on 14 degrees of freedom

Multiple R-squared: 0.9063, Adjusted R-squared: 0.8728

F-statistic: 27.08 on 5 and 14 DF, p-value: 9.927e-07

La matriz de diseno que debera recibir la implementacion manual es:

```
X <- model.matrix(modelo_ref)
y <- model.response(model.frame(modelo_ref))

n <- nrow(X)
p <- ncol(X)
df_res <- n - p

list(
  n = n,
  p = p,
  grados_libertad_residuales = df_res,
  nombres_columnas_X = colnames(X)
)
```

\$n

[1] 20

\$p

[1] 6

\$grados_libertad_residuales

[1] 14

\$nombres_columnas_X

[1] "(Intercept)" "x1" "x2" "x3" "x4"
[6] "x5"

head(X)

	(Intercept)	x1	x2	x3	x4	x5
1	1	3.83	28.87	7.20	26.6	6.19
2	1	2.89	20.10	-11.71	24.4	5.17
3	1	2.86	69.05	12.32	25.7	7.04
4	1	2.92	65.40	14.28	25.7	7.10
5	1	3.06	29.59	6.31	25.4	6.15
6	1	2.07	44.82	6.16	21.6	6.41

4 Coeficientes e Inferencia Individual

La implementacion manual debe reproducir:

$$\hat{\beta} = (X^T X)^{-1} X^T y.$$

La matriz de varianzas estimada es:

$$\widehat{\text{Var}}(\hat{\beta}) = \hat{\sigma}^2 (X^\top X)^{-1}, \quad \hat{\sigma}^2 = \frac{SSE}{n-p}.$$

Para cada coeficiente:

$$t_j = \frac{\hat{\beta}_j}{se(\hat{\beta}_j)}, \quad p_j = 2P(T_{n-p} \geq |t_j|).$$

```
coef_ref <- as.data.frame(summary(modelo_ref)$coefficients)
coef_ref$termino <- rownames(coef_ref)
rownames(coef_ref) <- NULL
coef_ref <- coef_ref[, c("termino", "Estimate", "Std. Error", "t value", "Pr(>|t|)")]
coef_ref
```

	termino	Estimate	Std. Error	t value	Pr(> t)
1	(Intercept)	19.94856503	13.62754835	1.4638411	1.653274e-01
2	x1	-1.79333327	1.23339617	-1.4539799	1.680013e-01
3	x2	0.04360156	0.05325887	0.8186723	4.266896e-01
4	x3	0.55576006	0.09295638	5.9787188	3.376177e-05
5	x4	1.11016839	0.43376789	2.5593605	2.270137e-02
6	x5	-1.81092194	2.02738916	-0.8932286	3.868318e-01

```
ic_ref <- as.data.frame(confint(modelo_ref, level = 0.95))
ic_ref$termino <- rownames(ic_ref)
rownames(ic_ref) <- NULL
ic_ref <- ic_ref[, c("termino", "2.5 %", "97.5 %")]
ic_ref
```

	termino	2.5 %	97.5 %
1	(Intercept)	-9.27961926	49.1767493
2	x1	-4.43870495	0.8520384
3	x2	-0.07062736	0.1578305
4	x3	0.35638845	0.7551317
5	x4	0.17982881	2.0405080
6	x5	-6.15923922	2.5373953

5 Valores Ajustados y Residuos

La implementacion manual debe reproducir:

$$\hat{y} = X\hat{\beta}, \quad e = y - \hat{y}.$$

```
resumen_obs <- data.frame(
  obs = seq_along(y),
  y = y,
  fitted = fitted(modelo_ref),
  residual = resid(modelo_ref),
  leverage = hatvalues(modelo_ref),
  std_residual = rstandard(modelo_ref),
  cooks_distance = cooks.distance(modelo_ref)
)

resumen_obs
```

	obs	y	fitted	residual	leverage	std_residual	cooks_distance
1	1	37.01	36.66122	0.34877932	0.4824874	0.23373274	0.0084889303
2	2	26.51	26.85992	-0.34991540	0.4862099	-0.23534201	0.0087354583
3	3	36.51	40.45972	-3.94972112	0.1330900	-2.04507245	0.1070132356
4	4	40.70	41.17361	-0.47360981	0.1711694	-0.25079397	0.0021649278
5	5	37.10	36.31909	0.78091123	0.1781281	0.41526840	0.0062292382
6	6	33.90	33.98570	-0.08569692	0.4999116	-0.05842129	0.0005686401
7	7	41.80	41.08124	0.71876050	0.2391022	0.39723756	0.0082643091
8	8	33.40	33.83425	-0.43425262	0.1074992	-0.22159880	0.0009857802
9	9	41.01	40.38558	0.62441635	0.2853502	0.35608763	0.0084381519
10	10	37.20	36.98960	0.21039502	0.6180439	0.16411867	0.0072639117
11	11	23.30	25.50804	-2.20804084	0.2914317	-1.26457768	0.1096212589
12	12	35.20	33.45353	1.74646934	0.4027088	1.08942465	0.1333669518
13	13	34.90	35.94872	-1.04872311	0.3693005	-0.63661796	0.0395515944
14	14	33.10	33.44567	-0.34566703	0.1088052	-0.17652284	0.0006340560
15	15	22.70	24.47888	-1.77888087	0.3464445	-1.06080337	0.0994190591
16	16	39.70	38.40258	1.29741721	0.1568976	0.68119077	0.0143920317
17	17	31.80	33.23983	-1.43983338	0.2906190	-0.82414143	0.0463763714
18	18	31.70	26.69824	5.00175644	0.3255630	2.93616941	0.6935927484
19	19	43.10	41.97738	1.12261550	0.2845680	0.63984697	0.0271405461
20	20	41.01	40.74718	0.26282017	0.2226699	0.14370942	0.0009859961

```
summary(resid(modelo_ref))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-3.94972	-0.61739	0.06235	0.00000	0.73430	5.00176

6 Sumas de Cuadrados

Con intercepto, usamos sumas de cuadrados corregidas por la media:

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2, \quad SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2.$$

Tambien:

$$SST = SSR + SSE.$$

```
y_bar <- mean(y)
y_hat <- fitted(modelo_ref)
e <- resid(modelo_ref)

SSE <- sum(e^2)
SST <- sum((y - y_bar)^2)
SSR <- sum((y_hat - y_bar)^2)

MSE <- SSE / df_res
MSR <- SSR / (p - 1)
sigma_hat <- sqrt(MSE)

ss_ref <- data.frame(
  cantidad = c("SST", "SSR", "SSE", "MSR", "MSE", "sigma_hat"),
  valor = c(SST, SSR, SSE, MSR, MSE, sigma_hat)
)

ss_ref
```

	cantidad	valor
1	SST	642.924375
2	SSR	582.686522
3	SSE	60.237853
4	MSR	116.537304
5	MSE	4.302704
6	sigma_hat	2.074296

```
data.frame(
  identidad = "SST - SSR - SSE",
  valor = SST - SSR - SSE
)
```

	identidad	valor
1	SST - SSR - SSE	-6.394885e-14

7 R Cuadrada y Error Estandar Residual

$$R^2 = 1 - \frac{SSE}{SST}, \quad R_{adj}^2 = 1 - \frac{SSE/(n-p)}{SST/(n-1)}.$$

```

R2 <- 1 - SSE / SST
R2_adj <- 1 - (SSE / df_res) / (SST / (n - 1))

data.frame(
  R2 = R2,
  R2_ajustada = R2_adj,
  sigma_hat = sigma_hat
)

```

```

      R2 R2_ajustada sigma_hat
1 0.9063065   0.8728445   2.074296

```

8 Prueba F Global

La prueba global contrasta:

$$H_0 : \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0.$$

contra la alternativa de que al menos un coeficiente de pendiente es distinto de cero.

$$F = \frac{SSR/(p-1)}{SSE/(n-p)} = \frac{MSR}{MSE}.$$

```

F_global <- MSR / MSE
p_F_global <- pf(F_global, df1 = p - 1, df2 = df_res, lower.tail = FALSE)

data.frame(
  estadistico_F = F_global,
  gl_numerador = p - 1,
  gl_denominador = df_res,
  p_value = p_F_global
)

```

```

      estadistico_F gl_numerador gl_denominador      p_value
1      27.08467           5           14 9.926837e-07

```

```
anova(modelo_ref)
```

Analysis of Variance Table

```

Response: y
      Df Sum Sq Mean Sq F value    Pr(>F)
x1      1  23.77    23.77    5.5251 0.03393 *

```

```

x2          1 343.23  343.23 79.7709 3.715e-07 ***
x3          1 186.34  186.34 43.3086 1.227e-05 ***
x4          1  25.91   25.91  6.0209  0.02784 *
x5          1   3.43    3.43  0.7979  0.38683
Residuals 14  60.24    4.30
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Nota: `anova(modelo_ref)` muestra sumas de cuadrados secuenciales. Para comparar la prueba global contra el modelo con solo intercepto, usar los calculos de SSR, SSE y `F_global` anteriores.

9 Devianza, Log-Verosimilitud, AIC y BIC

En el modelo gaussiano ajustado por minimos cuadrados, la devianza reportada por `deviance()` coincide con SSE.

La log-verosimilitud maximizada, usando el estimador de maxima verosimilitud $\hat{\sigma}_{ML}^2 = SSE/n$, es:

$$\ell(\hat{\beta}, \hat{\sigma}_{ML}^2) = -\frac{n}{2} \left[\log(2\pi) + 1 + \log\left(\frac{SSE}{n}\right) \right].$$

Para `AIC.lm`, R cuenta como parametros los p coeficientes y un parametro adicional de escala, por lo que $k = p + 1$.

```

sigma2_ML <- SSE / n
logLik_manual <- -n / 2 * (log(2 * pi) + 1 + log(sigma2_ML))
k <- p + 1

lik_ref <- data.frame(
  cantidad = c(
    "deviance_lm",
    "SSE",
    "sigma2_ML",
    "logLik_manual",
    "logLik_lm",
    "AIC_manual",
    "AIC_lm",
    "BIC_manual",
    "BIC_lm"
  ),
  valor = c(
    deviance(modelo_ref),
    SSE,
    sigma2_ML,
    logLik_manual,

```

```

    as.numeric(logLik(modelo_ref)),
    -2 * logLik_manual + 2 * k,
    AIC(modelo_ref),
    -2 * logLik_manual + log(n) * k,
    BIC(modelo_ref)
  )
)

lik_ref

```

	cantidad	valor
1	deviance_lm	60.237853
2	SSE	60.237853
3	sigma2_ML	3.011893
4	logLik_manual	-39.404457
5	logLik_lm	-39.404457
6	AIC_manual	92.808915
7	AIC_lm	92.808915
8	BIC_manual	99.779041
9	BIC_lm	99.779041

10 Modelo Reducido y Prueba F Parcial

Para tener una segunda referencia de comparacion, usamos un modelo reducido:

$$y_i = \beta_0 + \beta_3 x_{i3} + \beta_4 x_{i4} + \beta_5 x_{i5} + \varepsilon_i.$$

La hipotesis parcial es:

$$H_0 : \beta_1 = \beta_2 = 0 \mid x_3, x_4, x_5.$$

```

modelo_red <- lm(y ~ x3 + x4 + x5, data = datos)
summary(modelo_red)

```

Call:

```
lm(formula = y ~ x3 + x4 + x5, data = datos)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-3.8509	-1.1263	-0.1483	1.2599	4.7944

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)

```

(Intercept) 18.39794    12.53800    1.467    0.1617
x3           0.57485     0.08874    6.478 7.62e-06 ***
x4           0.74182     0.37586    1.974  0.0659 .
x5          -0.59432     1.29319   -0.460  0.6520
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Residual standard error: 2.114 on 16 degrees of freedom
Multiple R-squared: 0.8888, Adjusted R-squared: 0.868
F-statistic: 42.64 on 3 and 16 DF, p-value: 7.401e-08

```
anova(modelo_red, modelo_ref)
```

Analysis of Variance Table

```

Model 1: y ~ x3 + x4 + x5
Model 2: y ~ x1 + x2 + x3 + x4 + x5
  Res.Df  RSS Df Sum of Sq    F Pr(>F)
1      16 71.483
2      14 60.238  2    11.245 1.3067 0.3018

```

Manual:

$$F = \frac{(SSE_R - SSE_C)/(p_C - p_R)}{SSE_C/(n - p_C)}.$$

```

SSE_red <- sum(resid(modelo_red)^2)
p_red <- length(coef(modelo_red))

F_parcial <- ((SSE_red - SSE) / (p - p_red)) / (SSE / (n - p))
p_F_parcial <- pf(F_parcial, df1 = p - p_red, df2 = n - p, lower.tail = FALSE)

data.frame(
  SSE_reducido = SSE_red,
  SSE_completo = SSE,
  gl_numerador = p - p_red,
  gl_denominador = n - p,
  F_parcial = F_parcial,
  p_value = p_F_parcial
)

```

```

  SSE_reducido SSE_completo gl_numerador gl_denominador F_parcial  p_value
1      71.48276      60.23785           2           14  1.306726 0.3017707

```

11 Prueba de Razon de Verosimilitudes

Para el mismo contraste parcial, la prueba de razon de verosimilitudes usa:

$$LR = 2(\ell_C - \ell_R).$$

y se compara de forma asintotica con una χ^2 con $p_C - p_R$ grados de libertad.

```
LR <- 2 * (as.numeric(logLik(modelo_ref)) - as.numeric(logLik(modelo_red)))
p_LR <- pchisq(LR, df = p - p_red, lower.tail = FALSE)

data.frame(
  LR = LR,
  gl = p - p_red,
  p_value = p_LR
)
```

```
      LR gl  p_value
1 3.423108 2 0.1805849
```

12 Objetivos para la Implementacion Manual

La implementacion manual/C++ debera recibir, como minimo, una matriz **X** y un vector **y**. Para este documento, esa matriz tiene 20 filas y 6 columnas: intercepto, x1, x2, x3, x4, x5.

El primer nucleo debe reproducir:

- **beta_hat**: coeficientes estimados.
- **fitted**: valores ajustados.
- **residuals**: residuos ordinarios.
- SSE, SSR, SST.
- MSE, MSR, sigma_hat.
- R2 y R2_adj.
- **vcov(beta_hat)**.
- errores estandar, estadisticos t y p-values.
- intervalos de confianza.
- prueba F global.
- devianza gaussiana.
- log-verosimilitud, AIC y BIC.
- prueba F parcial entre modelo reducido y completo.
- LRT entre modelo reducido y completo.

13 Referencias Numericas Compactas

Esta ultima tabla deja los valores escalares principales en un formato facil de comparar.

```
referencia_escalar <- data.frame(
  cantidad = c(
    "n",
    "p",
    "df_res",
    "SST",
    "SSR",
    "SSE",
    "MSE",
    "MSR",
    "sigma_hat",
    "R2",
    "R2_adj",
    "F_global",
    "p_F_global",
    "logLik",
    "AIC",
    "BIC",
    "SSE_reducido",
    "F_parcial",
    "p_F_parcial",
    "LR",
    "p_LR"
  ),
  valor = c(
    n,
    p,
    df_res,
    SST,
    SSR,
    SSE,
    MSE,
    MSR,
    sigma_hat,
    R2,
    R2_adj,
    F_global,
    p_F_global,
    as.numeric(logLik(modelo_ref)),
    AIC(modelo_ref),
    BIC(modelo_ref),
    SSE_red,
    F_parcial,
```

```

    p_F_parcial,
    LR,
    p_LR
  )
)

referencia_escalara

```

	cantidad	valor
1	n	2.000000e+01
2	p	6.000000e+00
3	df_res	1.400000e+01
4	SST	6.429244e+02
5	SSR	5.826865e+02
6	SSE	6.023785e+01
7	MSE	4.302704e+00
8	MSR	1.165373e+02
9	sigma_hat	2.074296e+00
10	R2	9.063065e-01
11	R2_adj	8.728445e-01
12	F_global	2.708467e+01
13	p_F_global	9.926837e-07
14	logLik	-3.940446e+01
15	AIC	9.280891e+01
16	BIC	9.977904e+01
17	SSE_reducido	7.148276e+01
18	F_parcial	1.306726e+00
19	p_F_parcial	3.017707e-01
20	LR	3.423108e+00
21	p_LR	1.805849e-01